K-Nearest Neighbors (KNN)

- **Concept**: KNN is a non-parametric, instance-based learning algorithm that classifies data based on the majority label of the k-nearest data points in the feature space.
- **Intuition**: The assumption is that similar data points are near each other, so proximity is a good indicator for classification or regression.
- **Strengths**:
  - Simple to understand and implement.
  - No training phase; it's a lazy learner.
  - Works well with small datasets.
- **Weaknesses**:
  - Sensitive to noisy data and irrelevant features.
  - High computational cost at prediction time for large datasets.

Support Vector Machines (SVM)

- **Concept**: SVM is a supervised learning model that finds the optimal hyperplane that maximally separates data points from different classes in a high-dimensional space.
- **Intuition**: The model aims to create the widest margin between classes, using support vectors to define the decision boundary.
- **Strengths**:
  - Effective in high-dimensional spaces.
  - Robust to overfitting with proper kernel choice.
- **Weaknesses**:
  - Computationally intensive for large datasets.
  - Sensitive to choice of kernel and hyperparameters.

Decision Trees

- **Concept**: Decision Trees split the data recursively on feature thresholds, creating a tree-like structure for classification or regression.
- **Intuition**: They model decisions as a sequence of questions, splitting the data at each node to

increase purity.

- **Strengths**:
  - Easy to interpret and visualize.
  - Can handle both numerical and categorical data.
- **Weaknesses**:
  - Prone to overfitting, especially with deep trees.
  - Unstable to small data changes.

Ensemble Learning

- **Concept**: Combines multiple models (often weak learners like decision trees) to create a stronger, more robust model.
- **Intuition**: By aggregating the predictions of many models, variance can be reduced (Bagging) or bias can be reduced (Boosting).
- **Strengths**:
  - Improves performance over single models.
  - Reduces variance and/or bias.
- **Weaknesses**:
  - Less interpretable.
  - Higher computational cost.

Unsupervised Learning - Clustering

- **Concept**: Grouping data points into clusters based on similarity without using labels.
- **Intuition**: Clustering identifies natural groupings in data, like customer segments or anomaly detection.
- **Strengths**:
  - Reveals patterns and structure in unlabeled data.
  - Useful for exploratory analysis.
- **Weaknesses**:
  - Sensitive to choice of distance metric and number of clusters.
  - May not capture complex structures.

Singular Value Decomposition (SVD)

- **Concept**: Matrix factorization technique that decomposes a matrix into three matrices (U, Sigma, VT transpose) for dimensionality reduction or latent factor discovery.
- **Intuition**: Reduces complex data to a simpler form while preserving essential structure.
- **Strengths**:
  - Useful for dimensionality reduction and noise filtering.
  - Foundation of many recommendation systems.
- **Weaknesses**:
  - Computationally expensive for large matrices.
  - Can lose interpretability in reduced space.

Curse of Dimensionality

- **Concept**: As the number of features increases, the data becomes sparse, making learning difficult.
- **Intuition**: Higher dimensions dilute the density of data, requiring exponentially more samples.
- **Strengths**:
  - Highlights the importance of feature selection.
- **Weaknesses**:
  - Can lead to overfitting, poor generalization.
  - Distance measures lose meaning.

Principal Component Analysis (PCA)

- **Concept**: A linear transformation that projects data into a lower-dimensional space by maximizing variance along principal components.
- **Intuition**: Compresses data while retaining the most important information (variance).
- **Strengths**:
  - Reduces dimensionality for visualization and modeling.
  - Decorrelates features.
- **Weaknesses**:
  - Linear method; may not capture nonlinear patterns.

  - Components may not be easily interpretable.

Naive Bayes' Classifier

- **Concept**: A probabilistic classifier based on Bayes' theorem, assuming feature independence.
- **Intuition**: Estimates class probabilities based on independent feature likelihoods.
- **Strengths**:
  - Simple, fast, and effective for high-dimensional data.
  - Performs well with small datasets.
- **Weaknesses**:
  - Independence assumption is often unrealistic.
  - Sensitive to feature correlation.